

MTN Uganda Solution Description

Equity Bank Agent Float Liquidation (16924)



Copyright

© Ericsson AB 2019 - All Rights Reserved

Disclaimer

No part of this document may be reproduced in any form without the written permission of the copyright owner. The contents of this document are subject to revision without notice due to continued progress in methodology, design and manufacturing. Ericsson shall have no liability for any error or damage of any kind resulting from the use of this document.

Trademark List

Ericsson is the trademark or registered trademark of Telefonaktiebolaget LM Ericsson. All other products or service names mentioned in this document are trademarks of their respective companies

Contributors:

Name	Role
Beegira Atuhaire	Business Analyst

Revision History

Revision	Date	Comments
PA1	17-Feb-2022	Initial Draft



Contents

Copyright	2
1	Introduction5
1.1	Requirement5
1.2	Scope5
1.3	Intended Audience5
1.4	Prerequisites5
2	Implementation6
2.1	GetAccountHolderInfo6
2.2	Debit6
3	API Samples.....7
3.1	Gettransactionstatus7
3.2	GetAccountHolderInfo.....8
3.3	Debit Request.....9
3.4	DebitCompleted11
3.5	Getbalance (Optional)12
3.6	Refund (Optional).....13
4	Mobile Money Integration With MTN15
4.1	Connectivity15
4.2	Client Authentication15
4.3	Application layer – Normal Authentication.....15
4.4	Login Service16
4.5	Credential Information16
4.6	Application Layer - Basic Authentication17
4.7	Error Handling17
4.8	Protocol Structure.....18
4.9	URI Syntax18
4.10	Response Status Codes18
4.11	SSL Certificate Setup.....19
4.12	The SSL Handshake Process19
5	Security Description – CLM ID: 7124.....22
5.1	Security Impact22
5.2	Security Risk Management22
5.3	Security Requirement22
6	USSD Menu25
7	Exclusion.....26
8	Assumptions.....26



9	Impact	26
10	Notifications.....	26
10.1	SMS Notifications	26
10.2	USSD Notifications	26
10.3	XML Notifications.....	26
11	References	27
12	Glossary	27
13	Expected Test Cases.....	28
14	ECW Architectural Diagram.....	28
Approval		29
Appendix 1: Standard Error Codes		30



1 Introduction

1.1 Requirement

MTN Uganda is partnering with Equity bank to enable MTN Mobile Money agents to exchange their e-float for cash (cash-out service) at Equity branch and Agency points. Transactions will be initiated by the bank and will have to be approved by the agents on their own handsets. The following APIs shall be used:

- Debit
- DebitCompleted
- GetAccountHolderInfo
- Refund
- Getbalance
- GetTransactionStatus

1.2 Scope

The request solution design includes:

- Solution Design
- NAM Adaptation

1.3 Intended Audience

The following are included in the target group for this document:

- Solution Architects
- Account Managers
- Project Managers
- System Integrators

1.4 Prerequisites

- NAM must be ready to process the division code returned.
- The user should be in activated state.



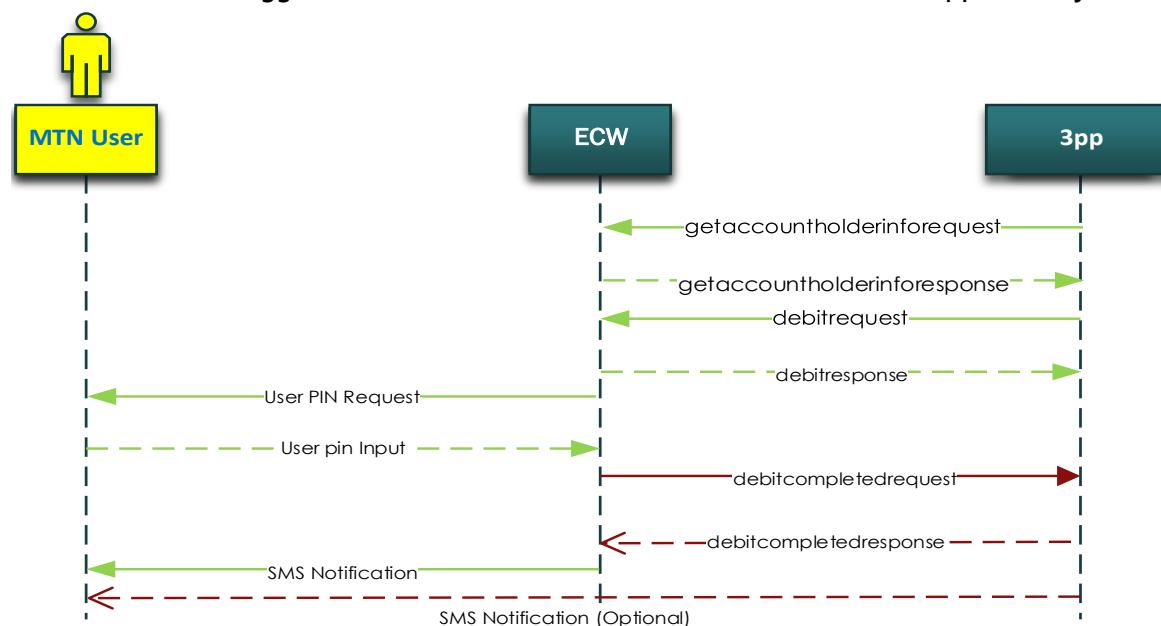
2 Implementation

2.1 GetAccountholderInfo

This API will be used by the 3PP (bank) to fetch recipient details (names) for validation before pushing a request. The details returned shall include recipient names and status at ECW.

2.2 Debit

This API shall be triggered from the bank's interface and will have to be approved by the agent.





3 API Samples

3.1 Gettransactionstatus

This API is used to query the status of get status of previously initiated Payment, Transfer or Debit transaction initiated by a 3PP. This API may be triggered in case a response for a completed request isn't received from the mobile money platform.

URI: `/{{baseContext}}/gettransactionstatus`

Namespace: http://www.ericsson.com/em/emm/financial/v1_1

Schema file: `gettransactionstatus.xsd`

8.5.1 Parameters

`gettransactionstatusrequest`:

Table 18 <code>gettransactionstatusrequest</code>			
Name	Type	Optionality	Description
<code>referenceid</code>	string	Optional	Reference ID which has been entered by user, while initiating the transaction. Validated with Length (min=1) (max=64). Validated with IsExternalReferenceString .
<code>financialtransactionid</code>	long	Optional	Financial ID of the transaction, received in the previously initiated operation.
<code>identity</code>	string	Optional	A valid identity associated with the account holder that performed the requested transaction. Validated with IsIdentity .

`gettransactionstatusresponse`:

Table 19 <code>gettransactionstatusresponse</code>			
Name	Type	Optionality	Description
<code>financialtransactionid</code>	long	Mandatory	The financial transaction ID. Parameter can not be NULL.
<code>status</code>	financialtransactionstatus v1.0	Mandatory	Payment Status. Parameter can not be NULL.

Sample Request



```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns2:gettransactionstatusrequest xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_0">
  <referenceid>900057716798</referenceid>
</ns2:gettransactionstatusrequest>
```

Sample response

```
<?xml version="1.0" encoding="UTF-8"?>
<ns2:gettransactionstatusresponse xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_0">
  <referenceid>900057716798</referenceid>
  <status>SUCCESSFUL</status>
</ns2:gettransactionstatusresponse>
```

Get Transaction Status Error Response

```
<ns2:errorResponse xmlns:ns2="http://www.ericsson.com/lwac" errorcode="TRANSACTION_NOT_FOUND"/>
```

3.2 GetAccountHolderInfo

Of particular interest to the 3PP should be the first name and surname, which will then be displayed to the subscriber to confirm the transaction.

URI: `/{{baseContext}}/getaccountholderinfo`

Namespace: http://www.ericsson.com/em/emm/provisioning/v1_1

Schema file: `getaccountholderinfo.xsd`

50.60.1 Parameters

getaccountholderinforequest:

Table 863 getaccountholderinforequest			
Name	Type	Optionality	Description
identity	string	Optional	A valid identity associated with the account holder. If omitted, the logged in user details are returned. Validated with IsIdentity .

getaccountholderinforesponse:

Table 864 getaccountholderinforesponse			
Name	Type	Optionality	Description



accountholderbasicinfo	accountholderbasicinfotype v1.1	Optional	Contains a subset of account holder information.
------------------------	------------------------------------	----------	--

Sample Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns0:getaccountholderinfo request xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_1">
  <identity>ID:256789999576/MSISDN</identity>
</ns0:getaccountholderinfo request>
```

Sample Response:

```
<?xml version="1.0" encoding="UTF-8"?>
<ns0:getaccountholderinfo response xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_1">
  <accountholderbasicinfo>
    <msisdn>256789999576</msisdn>
    <firstname>Emolut</firstname>
    <surname>Main</surname>
    <accountholderstatus>ACTIVE</accountholderstatus>
  </accountholderbasicinfo>
</ns0:getaccountholderinfo response>
```

3.3 Debit Request

A recipient account holder can request to debit another account holders account. The transfer to the recipient is completed when the account holder approves the debit.

URI: `/baseContext/debit`

Namespace: http://www.ericsson.com/em/emm/financial/v1_1

Schema file: debit.xsd

Parameters

debitrequest:			
Name	Type	Optionality	Description
fromfri	string	Mandatory	The FRI to identify the account holder from whom the money will be debited. This may identify an account holder, in such case the default account will be used for the Debit. Note: the account to debit must be a Mobile Money account. Parameter can not be NULL. Validated with IsFRI .



tofri	string	Mandatory	The FRI to identify the recipient to whom the money is credited. Parameter can not be NULL. Validated with IsFRI .
amount	moneydetailstype v1.0	Mandatory	The amount that is to be debited from the debtors account FRI to the creditors FRI. Parameter can not be NULL. Validated with IsPositiveAmount .
externaltransactionid	string	Mandatory	External system transaction ID. This ID will be associated with the created Debit transaction. Parameter can not be NULL. Validated with Length (min=1) (max=256). Validated with IsExternalReferenceString .
frommessage	string	Optional	A descriptive message for the debtors transaction history. If present, it will also be added to the sendacknowledgerequest and notification sent to the debtor. Validated with Length (max=256). Validated with IsRestrictedString .
tomessage	string	Optional	A descriptive message for the creditors transaction history. Validated with Length (max=256). Validated with IsRestrictedString .
referenceid	string	Optional	A optional unique external reference identity. This identity must be unique for each request for the logged in service provider. If no response is sent back to the service provider due to a communication error etc, then the service provider can use this identity to ask about the status of the transfer. Validated with Length (max=64). Validated with IsExternalReferenceString .
quoteid	quoteid v1.0	Optional	The ID of the quote to use.
coupon	coupon v1.0	Optional	A coupon the user would like to redeem and use the reward as part of this debit

debitresponse:

Table 11 debitresponse			
Name	Type	Optionality	Description
transactionid	long	Mandatory	The financial transaction ID. Parameter can not be NULL.
status	financialtransactionstatus v1.0	Mandatory	Debit status. Parameter can not be NULL.

Values to be signed: transactionid, status

Sample Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns2:debitrequest xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_1">
<fromfri>FRI:26878642189/MSISDN</fromfri>
```



```
<tofri>FRI:Equity-agent.sp1/USER</tofri>
<amount>
<amount>20</amount>
<currency>UGX</currency>
</amount>
<externaltransactionid>123ABCFD4</externaltransactionid>
<referenceid>1234567890</referenceid>
</ns2:debitrequest>
```

Sample Response:

```
<ns6:debitresponse xmlns:ns6="http://www.ericsson.com/em/emm/financial/v1_1"
xmlns:fic="http://www.ericsson.com/em/emm/financial/v1_0/common" xmlns:ns4="http://
www.ericsson.com/em/emm/coupons/v1_0/common"
xmlns:op="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <transactionid>26610501</transactionid>
  <status>PENDING</status>
</ns6:debitresponse>
```

The following status options apply:

Value	Description
COMPLETED	Operation completed.
CREATED	Operation created.
EXPIRED	Operation expired.
FAILED	Operation failed.
PENDING	Operation pending.

3.4 DebitCompleted

This API will be used to complete a push request when a user approves a transaction and will be sent by ECW to the 3PP.

--- sample request ----

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns0:debitcompletedrequest xmlns:ns0="http://www.ericsson.com/em/emm/financial/v1_1">
  <transactionid>2661050</transactionid>
  <externaltransactionid>445</externaltransactionid>
  <receiverinfo>
    <fri>FRI:Equity-agent.sp1/USER</fri>
  </receiverinfo>
```



```
<status> SUCCESSFUL </status>
</ns0:debitcompletedrequest>
```

Sample response

HTTP Response code: 200

3.5 Getbalance (Optional)

This is used to query the balance for a specific account.

URI: `/[baseContext]/getbalance`

Namespace: http://www.ericsson.com/em/emm/financial/v1_0

Schema file: `getbalance.xsd`

Parameters

getbalancerequest			
Name	Type	Optionality	Description
fri	string	Optional	Used for a specific account, for example commissioning account balance. Validated with IsFRI .
quoteid	quoteid v1.0	Optional	The ID of the quote to use.

getbalanceresponse

Table 329 getbalanceresponse			
Name	Type	Optionality	Description
balance	moneydetailstype v1.0	Optional	Account holder's available account balance. Validated with IsPositiveAmount .
loyalty	loyalty v1.0	Optional	Loyalty information generated by the transaction.

Sample request:

```
<ns2:getbalancerequest xmlns:ns2=" http://www.ericsson.com/em/emm/financial/v1_1">
</ns2:getbalancerequest>
```

Sample response:

```
<ns2:getbalanceresponse xmlns:ns2=" http://www.ericsson.com/em/emm/financial/v1_1">
<balance>
```



```

<amount>
  <amount>500.00</amount>
  <currency>UGX</currency>
</amount>
</balance>
</ns2:getbalanceresponse>

```

3.6 Refund (Optional)

Refund money or loyalty points to an account holder. Bank may use this in case where agent does not receive actual cash for a transaction that was successful on the interface and at ECW.

URI: `/[baseContext]/refund`

Namespace: http://www.ericsson.com/em/emm/financial/v1_0

Schema file: refund.xsd

Parameters

refundrequest			
Name	Type	Optionality	Description
amount	moneydetailstype v1.0	Mandatory	The amount to refund. Parameter can not be NULL. Validated with IsPositiveAmount .
refundernote	string	Optional	A descriptive note for sender transaction history. Validated with Length (max=256). Validated with IsRestrictedString .
receivermessage	string	Optional	A descriptive note for receiver transaction history. Validated with Length (max=256). Validated with IsRestrictedString .
financialtransactionid	long	Mandatory	Transaction ID of the debit or payment transaction for which refund is requested. Validated with IsPositiveLong (allowNull=false) (allowZero=false).
quoteid	quoteid v1.0	Optional	The ID of the quote to use.

refundresponse

Table 541 refundresponse			
Name	Type	Optionality	Description
financialtransactionid	long	Mandatory	The financial transaction ID of the transfer. Validated with IsPositiveLong (allowNull=false) (allowZero=false).
refunderbalanceafter	moneydetailstype v1.0	Optional	Balance of the refunder account after the transaction. Validated with IsPositiveAmount .
refunderfee	moneydetailstype v1.0	Optional	The fee paid by the sender, currency is the same as the senders account defined by the sender FRI. Validated with IsPositiveAmount .
loyalty	loyalty v1.0	Optional	Loyalty information generated.

Sample request:



```
<ns0:refundrequest xmlns:ns0="http://www.ericsson.com/em/emm/financial/v1_0">
  <amount>
    <amount>1000.00</amount>
    <currency>UGX</currency>
  </amount>
  <refundernote></refundernote>
  <financialtransactionid>2661050</financialtransactionid>
  <receivermessage></receivermessage>
</ns0:refundrequest>
```

Sample response

```
<ns0:refundresponse xmlns:ns0="http://www.ericsson.com/em/emm/financial/v1_0">
  <refunderbalanceafter>
    <amount>
      <amount>1000.00</amount>
      <currency>UGX</currency>
    </amount>
  </refunderbalanceafter>
  <refunderfee>
    <amount>
      <amount>1000.00</amount>
      <currency>UGX</currency>
    </amount>
  </refunderfee>
  <financialtransactionid>2661050</financialtransactionid>
  <receivermessage></receivermessage>
</ns0:refundresponse>
```



4 Mobile Money Integration With MTN

4.1 Connectivity

The below URLs will be used for integration into the Mobile Money system, this will be done through a VPN. The URI will depend on the API and is similar to the API request, e.g. getbalancerequest the URI becomes getbalance, debitrequest becomes debit

Test Platform:

<https://10.156.145.219:8017/poextvip/v1/<uri>>

Production Primary Side:

<https://10.156.144.21:8006/poextvip/v1/<uri>>

Production Geo Side:

<https://10.156.191.21:8006/poextvip/v1/<uri>>

Outbound traffic: All traffic from the mobile money platform to the 3pp will also have a URI appended at the end of the URL, if a 3PP has a URL like <https://3ppendpoint.com/> and we are sending a payment or getfinancialresource information request to them, the final end point will be <https://3ppendpoint.com/deposit> or <https://3ppendpoint.com/paymentinstruction>

4.2 Client Authentication

The following authentication methods are supported:

Presentation layer: Server-side certificate

Authentication of the Mobile Money Manager service is supported through an HTTPS server certificate. In order to successfully connect to the interface, this certificate has to be trusted by the client, either directly, or through a certificate trust chain.

No matter if server-side certificates are used or not, the Application (the Service Provider system or the Service Bus) still needs to authenticate itself with a user name and a password. There are two methods supported in the Mobile Money Manager for this.

4.3 Application layer – Normal Authentication



Authentication of clients of the service is performed by a user name and password-based login service, described in 4.4.

The system returns the following in the HTTP header for the response:

Set-Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

The session ID is then used as a cookie in the HTTP header in the following requests towards the Mobile Money Manager.

Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

4.4 Login Service

A successful login to the system creates a session ID. The session ID can be used for multiple operations and will be kept alive and have its lifetime extended as long as operations are performed within the session lifetime.

Identifying oneself to the system by entering one's credential.

loginrequest:

loginrequest			
Name	Type	Multiplicity	Description
identity	String	Mandatory	Identity of user.
credential	credential information	Mandatory	Credential used to verify the users identity.

loginresponse:

An empty response. If HTTP response code 200 is returned the login is successful. If HTTP response code 302 is returned either a secondarylogin with an OTP or a changepassword are expected is needed before successful login. Which one is indicated in the response header.

4.5 Credential Information

edential			
Name	Type	Multiplicity	Description



secret	string	Mandatory	Credential secret
type	string	Mandatory	Credential secret format. Must be set to string.

4.6 Application Layer - Basic Authentication

The protocol also supports basic authentication scheme as defined in RFC 1945. This can be used instead of session establishment for performing an operation without requiring login.

The credentials need to be passed in the HTTP header in the following format:

Authorization: Basic QWxhZGRpbjpvGVuIHNIc2FtZQ==

Where the credentials are base64 encoded and in the format <login>:<password>.

4.7 Error Handling

When an operation results in an operation error, the HTTP connection will respond with the code 500. The operation errors may for example be invalid permissions, invalid parameter values or internal errors of the system. If response code 500 is returned, it is possible to unmarshall the `errorresponsetype` object from the error stream of the connection.

The actual error codes are subject to change, so they are currently not included in the description of each operation. Generally, the text for an error code allows a person to interpret the error correctly.

For example, the code `ACCOUNTHOLDER_NOT_FOUND` indicates that an identity was provided that does not have a match in the system.

The error code `RESOURCE_NOT_FOUND` similarly indicates that the requested financial resource, such as an account in the specified currency, does not exist in the system.

Note:

Prevent leaking too much information to an end user of the system. For example, when requesting a voucher that does not exist, the error `VOUCHER_NOT_FOUND` is returned. When requesting a voucher that exists but with another owner, the error `PERMISSION_DENIED` is returned. These two errors must not be displayed to an end



user since they expose a possibility to use brute force to get hold of valid voucher tokens in the system.

Below is the table for error handling:

errorResponse			
Name	Type	Multiplicity	Description
errorcode	xs:string	Mandatory	Refer to future list of error codes.
arguments	tns:errorargumenttype	Optional, 0..N	See errorArguments.
errorArgument			
name	xs:string	Mandatory	Additional info depending on error code.
value	xs:string	Mandatory	Additional value

4.8 Protocol Structure

This section describes the structure of the protocol.

4.9 URI Syntax

The URI of the HTTP request is structured according to the following pattern:

URI:= /{context}/{method} where context - is a fixed string representing the routing entity e.g. /poextvip method is a fixed string representing the operation type like debit,payment,spttransfer e.t.c. The same method can be used by different users. Depending on the user's authentication, a different context is used.

4.10 Response Status Codes

The protocol uses the regular HTTP status codes/ Status codes in the 200 range imply that the request completed normally. In this case the response body will be formatted according to the expected response syntax for the specified method. Status code in the 300 range indicate that the request must be resent to a different location. Status codes in the 400 or 500 ranges imply that there was an error executing the request.

The protocol uses the regular HTTP status codes/ Status codes in the 200 range imply that the request completed normally. In this case the response body will be formatted according to the expected response syntax for the specified method. Status code in the 300 range indicate that the request must be resent to



<i>a different location. Status codes in the 400 or 500 ranges imply that there was an error executing the request. Used HTTP response codes</i>	
<i>Response Code</i>	<i>Description</i>
<i>200 (OK)</i>	<i>Sent when the request completed successfully.</i>
<i>202 (Accepted)</i>	<i>The request has been accepted for processing, but the processing has not been completed (pending).</i>
<i>302 (Moved Temporarily)</i>	<i>The URL provided in the Location header field indicates where to send the following request and which request type to use. The following request types are possible:</i>
<i>401 (Unauthorized)</i>	<i>The request needs authentication.</i>
<i>403 (Forbidden)</i>	<i>The request was denied and will be denied also in the future.</i>
<i>404 (Not found)</i>	<i>The resource specified in the URI was not found.</i>
<i>500 (Internal Server Error)</i>	<i>The server encountered a general error during execution, see Section 57.</i>

4.11 SSL Certificate Setup

Follow the following steps for TLS v1.2 security setup, this is a guideline is used to ensure we have TLS v1.2 setup, we use the Mobile Money certificate authority to sign the certificates of the entity integrating to the Mobile Money Platform.

A 3rd party creates a TLS v1.2 certificate signing request (CSR) that is to be signed by the Mobile Money Platform.

- The Mobile Money Platform signs that TLS CSR with Mobile Money CA and share certificate as well as CA file with 3PP
- 3PP needs to import that certificate and CA in their keystore as trusted ones
- TLS v1.2 should be used
- On the mobile money platform, mutual authentication will be enabled to ensure only trusted entities are allowed

** Certificate renewal tracking is the responsibility of the 3PP.

4.12 The SSL Handshake Process

The easiest way to understand, interpret and troubleshoot SSL handshaking is through taking traces (**tcpdump**), and analysing the captured trace by using basic tools such as Wireshark. The snapshot below highlights the basic components of this process.



No.	Time	Source	Destination	Protocol	Length	Info
1	2020-01-31 17:09:06.365188	00:00:00_00:00...	00:00:00_00:00...	FILEINFO	177	tcpdump -nni any -so -vvv -w mag2.pcap host 10.135.103.1 and port 443
2	2020-01-31 17:11:19.563484	10.156.145.219	10.135.103.1	TCP	112	11021 → 443 [SYN] Seq=0 Win=4380 Len=0 MSS=1460 TSval=4226563667 TSecr=0 SACK_PERM=1
3	2020-01-31 17:11:19.764568	10.135.103.1	10.156.145.219	TCP	108	443 → 11021 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1380 SACK_PERM=1 TSval=14411824 TSecr=4226563667
4	2020-01-31 17:11:19.764602	10.156.145.219	10.135.103.1	TCP	104	11021 → 443 [ACK] Seq=1 Ack=1 Win=4380 Len=0 TSval=4226563868 TSecr=14411824
5	2020-01-31 17:11:19.764630	10.156.145.219	10.135.103.1	TLSv1.2	228	Client Hello
6	2020-01-31 17:11:19.965485	10.135.103.1	10.156.145.219	TCP	104	443 → 11021 [ACK] Seq=1 Ack=125 Win=28960 Len=0 TSval=14412027 TSecr=4226563868
7	2020-01-31 17:11:19.966625	10.135.103.1	10.156.145.219	TLSv1.2	1472	Server Hello
8	2020-01-31 17:11:19.966742	10.135.103.1	10.156.145.219	TCP	1472	443 → 11021 [PSH, ACK] Seq=1369 Ack=125 Win=28960 Len=1368 TSval=14412027 TSecr=4226563868 [TCP se
9	2020-01-31 17:11:19.966764	10.135.103.1	10.156.145.219	TCP	1464	443 → 11021 [PSH, ACK] Seq=2737 Ack=125 Win=28960 Len=1360 TSval=14412027 TSecr=4226563868 [TCP se
10	2020-01-31 17:11:19.966779	10.156.145.219	10.135.103.1	TCP	104	11021 → 443 [ACK] Seq=125 Ack=2737 Win=7116 Len=0 TSval=4226564071 TSecr=14412027
11	2020-01-31 17:11:19.966789	10.156.145.219	10.135.103.1	TCP	104	11021 → 443 [ACK] Seq=125 Ack=4097 Win=8476 Len=0 TSval=4226564071 TSecr=14412027
12	2020-01-31 17:11:20.167693	10.135.103.1	10.156.145.219	TLSv1.2	986	Certificate, Server Key Exchange, Server Hello Done

Figure 1: TCP Dump showing SSL handshake attempt with Client-Service Certificate.

No.	Time	Source	Destination	Protocol	Length	Info
5	2019-12-23 14:39:43.463792	10.156.145.219	10.156.62.133	TLSv1.2	221	Client Hello [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
6	2019-12-23 14:39:43.464331	10.156.62.133	10.156.145.219	TCP	97	443 → 52294 [ACK] Seq=1 Ack=125 Win=28960 Len=0 TSval=596734598 TSecr=847867568 [ETHERE
7	2019-12-23 14:39:43.466449	10.156.62.133	10.156.145.219	TLSv1.2	1465	Server Hello [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
8	2019-12-23 14:39:43.466999	10.156.62.133	10.156.145.219	TLSv1.2	831	Certificate, Server Key Exchange, Server Hello Done [ETHERNET FRAME CHECK SEQUENCE INCORRE
9	2019-12-23 14:39:43.467462	10.156.145.219	10.156.62.133	TCP	97	52294 → 443 [ACK] Seq=125 Ack=2103 Win=6482 Len=0 TSval=847867571 TSecr=596734600 [ETHERE
10	2019-12-23 14:39:43.471046	10.156.145.219	10.156.62.133	TLSv1.2	178	Client Key Exchange, Change Cipher Spec [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
11	2019-12-23 14:39:43.511012	10.156.62.133	10.156.145.219	TCP	97	443 → 52294 [ACK] Seq=2103 Ack=206 Win=28960 Len=0 TSval=596734645 TSecr=847867575 [ETHE
12	2019-12-23 14:39:43.511042	10.156.145.219	10.156.62.133	TLSv1.2	142	Encrypted Handshake Message [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
13	2019-12-23 14:39:43.511581	10.156.62.133	10.156.145.219	TCP	97	443 → 52294 [ACK] Seq=2103 Ack=251 Win=28960 Len=0 TSval=596734645 TSecr=847867615 [ETHE
14	2019-12-23 14:39:43.511589	10.156.62.133	10.156.145.219	TLSv1.2	148	Change Cipher Spec, Encrypted Handshake Message [ETHERNET FRAME CHECK SEQUENCE INCORRE
15	2019-12-23 14:39:43.511629	10.156.145.219	10.156.62.133	TCP	97	52294 → 443 [ACK] Seq=251 Ack=2154 Win=6533 Len=0 TSval=847867615 TSecr=596734645 [ETHERE
16	2019-12-23 14:39:43.512244	10.156.145.219	10.156.62.133	TCP	97	[TCP Dup ACK 15#1] 52294 → 443 [ACK] Seq=251 Ack=2154 Win=6533 Len=0 TSval=847867616 TSecr=
17	2019-12-23 14:39:43.512347	10.156.145.219	10.156.62.133	TLSv1.2	783	Application Data [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
18	2019-12-23 14:39:43.521372	10.156.62.133	10.156.145.219	TLSv1.2	332	Application Data [ETHERNET FRAME CHECK SEQUENCE INCORRECT]
19	2019-12-23 14:39:43.521412	10.156.145.219	10.156.62.133	TCP	97	52294 → 443 [ACK] Seq=937 Ack=2389 Win=6768 Len=0 TSval=847867625 TSecr=596734655 [ETHERE

Figure 2: TCP Dump showing successful SSL handshake attempt with client-service certificate (with application data).

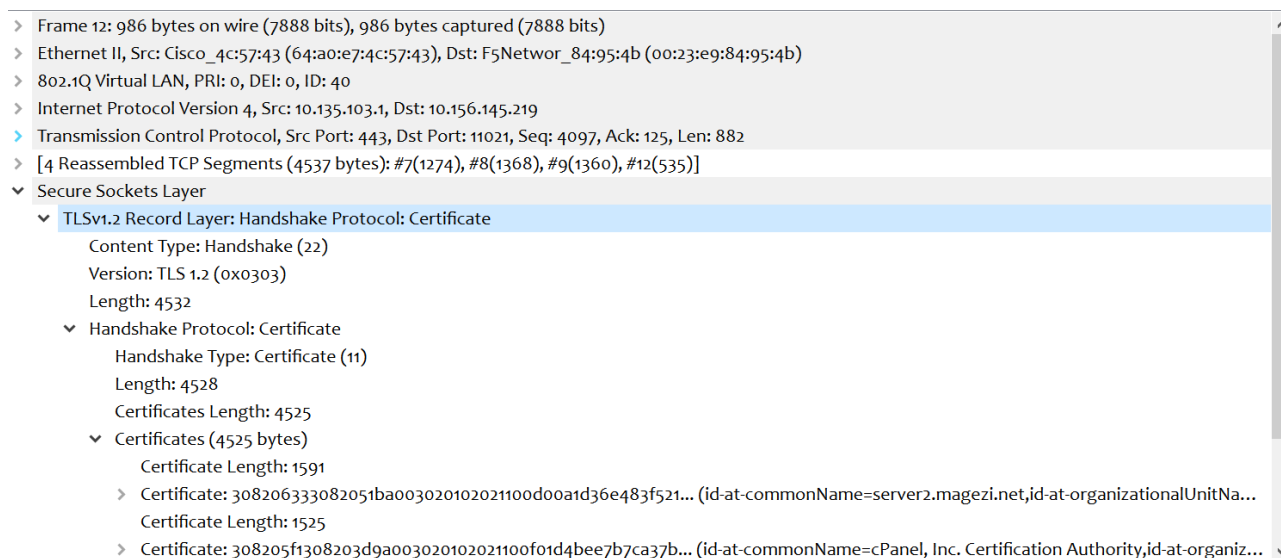
The exact steps within a TLS handshake will vary depending upon the kind of key exchange algorithm used and the cipher suites supported by both sides. The RSA key exchange algorithm is used most often. It goes as follows, in a typical 3PP→ECW handshake flow;

- 3PP (client) initiates a **SYN** request to the server
- Server responds with the **SYN, ACK**
- Client sends an **ACK** to confirm receipt of receipt of the packet.
- The '**client hello**' message: The client initiates the handshake by sending a "hello" message to the server. The message will include which TLS version the client supports, the cipher suites supported, and a string of random bytes known as the "client random."
- The '**server hello**' message: In reply to the client hello message, the server sends a message containing the server's SSL certificate, the server's chosen cipher suite, and the "server random," another random string of bytes that's generated by the server.
- Authentication: The client verifies the server's SSL certificate with the certificate authority that issued it. This confirms that the server is who it says it is, and that the client is interacting with the actual owner of the domain.
 - *The premaster secret: The client sends one more random string of bytes, the "premaster secret." The premaster secret is encrypted with the public key and can only be decrypted with the private key by the server. (The client gets the public key from the server's SSL certificate.)*
 - *Private key used: The server decrypts the premaster secret.*
 - *Session keys created: Both client and server generate session keys from the client random, the server random, and the premaster secret. They should arrive at the same results.*

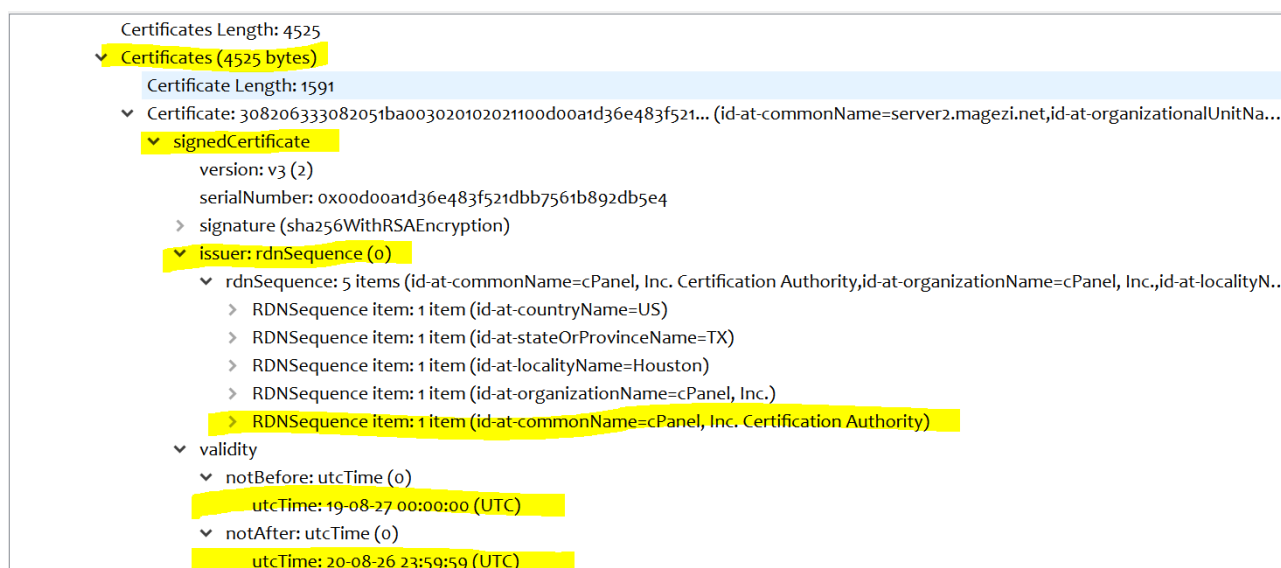


- Client is ready: The client sends a "finished" message that is encrypted with a session key.
- Server is ready: The server sends a "finished" message encrypted with a session key.
- Secure symmetric encryption achieved: The handshake is completed, and communication continues using the session keys.

The screenshot below shows the certificate chain being presented by the 3PP. In a standard ECW integration, only certificates signed by the ECW CA are permitted to complete a successful TLS handshake.



The screenshot below shows details of the issuer of the above certificate. In this example, the signer – cpanel. Such a handshake would be unsuccessful.





5 Security Description (CLM 16924)

5.1 Security Impact

This demand involves an integration an external system. The bank/3PP shall be expected to adhere to the baseline security guidelines and requirements, details of which have been outlined in the section below.

5.2 Security Risk Management

S/N	Risk Description	Rating	Vulnerability	Affected Component	Mitigation
			•		
			•		

5.3 Security Requirement

The security guidelines for this demand aligns with E// security standard. All existing security requirements applies.

Security Area	Domain	Description	Current Configuration/Evidence
Client Authentication	Access Control	Either of following should be supported at minimum for client authentication. - normal authentication (API session based). Authentication of clients is performed by a username/password-based API login service - via the basic authentication scheme as defined in RFC 1945	Username and password with basic auth
Mutual Authentication support	Access Control	TLS v1.2 based Mutual authentication must be supported i.e. Both Client and server must authenticate each other by presenting trusted certificates before	Mutual Authentication on TLS v1.2



Security Area	Domain	Description	Current Configuration/Evidence
		allowing any network connection	
Integrity Protection for Data in transit	Data protection	Software shall support message integrity through digital signature verification and creation. The signature algorithm such as RSA encryption over a SHA-256 digest of the signed data must be supported.	No data exchange outside the normal querying over TLS.
Strong encryption methods	Data Protection	If encryption is used for securing the data (rest, or in motion), the key lengths and algorithms shall be "strong" by current definitions. Any weak or broken encryption methods should not be used such as DES, MD5, etc	Existing EWP SoC application guidelines
Confidentiality & Integrity of data in motion	Network Data Protection	All communication interfaces internal or external must support secure protocols such as TLS (TLS 1.2 or above) to protect confidentiality and integrity of the messages.	Supported
Certificates issued by a CA	Network Data Protection	System must support use of X509 certificates for TLS based connectivity issued by an external certificate authority (CA preference must be agreed with operator). External CA must be trusted by system.	CA must be either M3 (ECW) or MTN. Existing certificates (M3 CA) shall be used.
Static IP addresses	Network Data Protection	The application should support use of static IP's for communication. If there is a strong need for dynamic IP's for communication, domain name resolution must be done only against an authorized domain	Supported



Security Area	Domain	Description	Current Configuration/Evidence
		names/system with support for DNSSEC.	
List of APIs.			• Debit • DebitCompleted • GetAccountHolderInfo • Refund • GetTransactionStatus
List of profiles and permissions			No new permissions required



6 USSD Menu

- N/A. No USSD menu required for this integration.
- 3PP Menu shall be out of scope.



7 Exclusion

The solution does not cover any other menu and customer journey developments at the bank side.

8 Assumptions

- The existing Architecture between ECW and 3pp will not change.
- The SSL VPN will still be maintained.
- There will be no change on all In-Bound USSD requests coming from 3pp to ECW.
- The only change required from ECW is to return a Response Code Reference for every menu Item that needs redirect.
- Once a Menu is redirected the Existing session to ECW is terminated.
- All 3rd Party interaction will be done on the 3PP menu system.
- 3PP will then send a debit request to ECW.
- Subscriber will use USSD Push to approve the debit.

9 Impact

This section covers the impact on existing functionality

10 Notifications

10.1 SMS Notifications

Existing standard notifications shall be used.

10.2 USSD Notifications

Existing USSD flash notifications shall be used.

10.3 XML Notifications

N/A



11 References

[1] URS

12 Glossary

3PP = Third Party Product

ECW = Ericsson Converged Wallet (Also the mobile money system)

PO = Partner Orchestrator (used as a gateway into Mobile Money)

XML = Extensible Mark Up Language

USSD = Unstructured Supplementary Service Data

SMS = Short Messaging Service

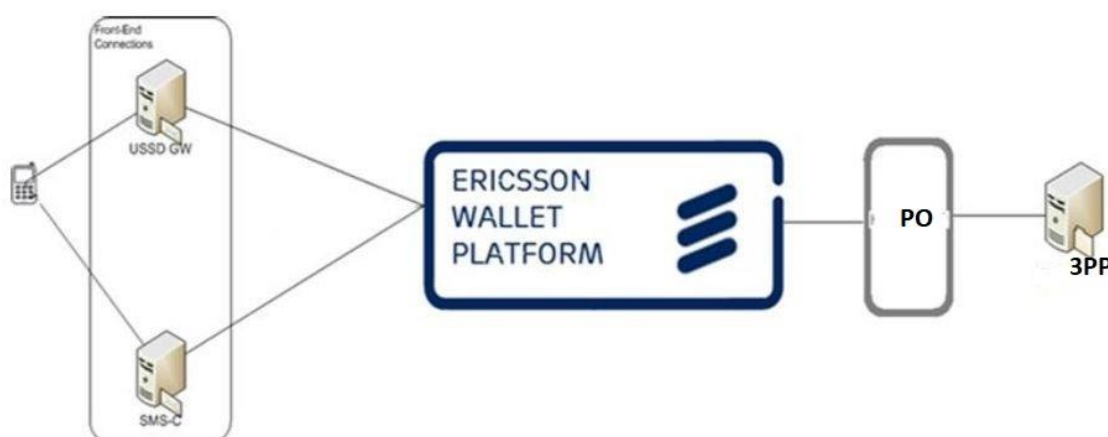
SSL = Secure Socket Layer



13 Expected Test Cases

S/N	Test Case	Expected Result
01	GetTransactionstatus	SUCCESSFUL
02	Debit	SUCCESSFUL
03	DebitCompleted	SUCCESSFUL

14 ECW Architectural Diagram





Approval

This document has been mutually reviewed and approved to constitute the full scope of the solution to be delivered by MTN Uganda and the 3PP. This document is signed in two (2) copies of which the parties have taken one (1) each.

MTN Uganda

Name	Role	Signature	Date
Abdul Ssebagala	Business Analyst, Planning and Architecture		
Daniel Tamale	Application Specialist		

Ericsson

Name	Role	Signature	Date
Beegira Atuhair	Business Analyst		
Agaba Simon Peter	Security Architect		



Appendix 1: Standard Error Codes

Error Code	Description
ACCOUNT_ALREADY_MEMBER_OF_GROUP	This account is already a member of this group
ACCOUNT_COULD_NOT_BE_REMOVED_FROM_BANKDOMAIN	Account could not be removed from Bank domain
ACCOUNT_HOLDER_ADDITIONAL_INFORMATION_NAME_ALREADY_EXISTS	The specified account holder's additional information name already exists
ACCOUNT_HOLDER_ADDITIONAL_INFORMATION_NAME_IN_USE	The specified account holder's additional information name cannot be deleted because it is in use
ACCOUNT_HOLDER_ALREADY_MEMBER_OF_GROUP	This account holder is already a member of this group
ACCOUNT_HOLDER_DOES_NOT_EXIST	AccountHolder does not exist
ACCOUNT_HOLDER_IS_ADMIN	The given action cannot be done to a group administrator.
ACCOUNT_HOLDER_IS_NOT_ADMIN	Only group admin of the group is allowed to perform the given action.
ACCOUNT_HOLDER_IS_NOT_MEMBER_OF_THE_GROUP	The given account holder is not a member of the group.
ACCOUNT_HOLDER_IS_SECRETARY	The given action cannot be done to a group secretary.
ACCOUNT_HOLDER_IS_TREASURER	The given action cannot be done to a group treasurer.
ACCOUNT_HOLDER_NOT_LOGGED_IN	This operation requires the user to be logged in
ACCOUNT_HOLDER_PROFILE_DOES_NOT_HAVE_REQUESTED_ACCOUNT_TYPE	AccountHolder profile does not have requested account
ACCOUNT_HOLDER_STATUS_IS_NOT_VALID_FOR_ADD_ACCOUNT_OPERATION	AccountHolder status should be blocked or active
ACCOUNT_NOT_FOUND	Account not found
ACCOUNT_ROUTE_ALREADY_EXISTS	Could not create account route since Account route with provided account route reference and currency already exists
ACCOUNT_ROUTE_NOT_FOUND	Could not delete account route since Account route with provided account route reference and currency does not exists
ACCOUNT_TRANSACTION_STATISTICS_COLLECTION_NOT_ALLOWED	Account transaction statistics collection not allowed on the account
ACCOUNT_TRANSACTION_STATISTICS_COLLECTION_NOT_ENABLED	Account transaction statistics collection not enabled on the account
ACCOUNT_TYPE_NOT_SUPPORTED	Account type is not supported
ACCOUNTHOLDER_ACCOUNT_REFERENCE_ALREADY_EXISTS	The account holder already has a connection to the specified account
ACCOUNTHOLDER_ACTIVATION_FAILED	Activation of account holder failed



ACCOUNTHOLDER_ANY_ONE_MANDATORY_FIELD_REQUIRED	Could not perform operation since at least one of the mandatory field is required
ACCOUNTHOLDER_CAN_NOT_REMOVE_EMAIL	Could not perform operation since at least one of the mandatory field is required
ACCOUNTHOLDER_CAN_NOT_REMOVE_EXT_ID	Could not perform operation since at least one of the mandatory field is required
ACCOUNTHOLDER_CAN_NOT_REMOVE_MSISDN	Could not perform operation since at least one of the mandatory field is required
ACCOUNTHOLDER_CAN_NOT_REMOVE_USER_NAME	Could not perform operation since at least one of the mandatory field is required
ACCOUNTHOLDER_EMAIL_ADDRESS_MISSING	Unable to send email to the accountholder because email address is missing
ACCOUNTHOLDER_EUI_NOT_FOUND	Account holder could not be found
ACCOUNTHOLDER_MSISDN_NOT_FOUND	Operation requires account holder to have an MSISDN identity but it has none
ACCOUNTHOLDER_NOT_ACTIVE	Account holder is inactive
ACCOUNTHOLDER_NOT_CHILD	A specified account holder was not a child to another specified account holder
ACCOUNTHOLDER_NOT_CONNECTED_TO_ACCOUNT	Account holder is not connected to the account
ACCOUNTHOLDER_NOT_FOUND	Account holder could not be found
ACCOUNTHOLDER_SETTINGS_ALREADY_EXISTS	The settings for this account holder already exists
ACCOUNTHOLDER_SETTINGS_MANDATORY	Account holder settings are mandatory
ACCOUNTHOLDER_SETTINGS_NOT_FOUND	Account holder settings not found
ACCOUNTHOLDER_WITH_ALIAS_ALREADY_EXISTS	Account holder could not be registered because an account holder with submitted Alias already exists
ACCOUNTHOLDER_WITH_EMAIL_ALREADY_EXISTS	Account holder could not be registered because an account holder with submitted E-mail already exists
ACCOUNTHOLDER_WITH_EMAIL_NOT_FOUND	Account holder with given e-mail could not be found
ACCOUNTHOLDER_WITH_EMPLOYEE_ID_ALREADY_EXISTS	Account holder cannot have the same employee id as another employee working under the same employer
ACCOUNTHOLDER_WITH_EXTERNALID_ALREADY_EXISTS	Account holder could not be registered because an account holder with submitted External ID already exists
ACCOUNTHOLDER_WITH_FRI_NOT_FOUND	Account holder with given FRI could not be found
ACCOUNTHOLDER_WITH_IMSI_ALREADY_EXISTS	Account holder IMSI could not be updated because it already exists
ACCOUNTHOLDER_WITH_MSISDN_ALREADY_EXISTS	Account holder could not be registered because an account holder with submitted MSISDN already exists
ACCOUNTHOLDER_WITH_MSISDN_NOT_FOUND	Account holder with given MSISDN could not be found



ACCOUNTHOLDER_WITH_USERNAME_ALREADY_EXISTS	Account holder could not be registered because an account holder with submitted Username already exists
ADJUSTMENTS_FROM_ACCOUNT_NOT_ALLOWED	Adjustments from account not allowed
ADJUSTMENTS_NOT_ALLOWED	Adjustments not allowed on account
ADJUSTMENTS_TO_ACCOUNT_NOT_ALLOWED	Adjustments to account not allowed
ALIAS_NOT_FOUND	No Alias with this name was found in the system
ALREADY_MEMBER_IN_ANOTHER_BANKDOMAIN	Could not perform operation since the resource already belongs to another Bank domain
AMBIGUOUS_CURRENCY	The FRI refers to more than one account
AMBIGUOUS_REFERENCE_ID	Reference id refers to more than one invoice.
AMOUNT_INVALID	Either amount and currency are incompatible, or currency is invalid
AMOUNT_TYPE_MISSING	Amount type is missing in scheduled transaction
API_GATEWAY_USER_ALREADY_EXISTS	API gateway user already registered for partner
API_GATEWAY_USER_NOT_FOUND	No API gateway user with given identity was found
APPROVER_ALREADY_HAS_PENDING_SWITCH_PAYOUT_CYCLE	The approver of switching a payout cycle with another contributor already has a pending switch.
APPROVER_IS_NOT_ACCEPTED_CONTRIBUTOR_IN_THE_GROUP	The approver of switching a payout cycle with another contributor has not accepted their invitation to the group.
AUTHORIZATION_ACCOUNT_NOT_ALLOWED_TO_PERFORM_ACTION	Account holder is not allowed to perform this operation
AUTHORIZATION_ACCOUNTHOLDER_ALREADY_ACTIVATED	Account holder is already activated
AUTHORIZATION_ACCOUNTHOLDER_NOT_ACTIVE	Account holder is not active
AUTHORIZATION_CURRENT_BALANCE_TOO_LOW	Insufficient amount
AUTHORIZATION_FAILED	Authorization failed
AUTHORIZATION_INVALID_ACCOUNT_TYPE_COMBINATION	Performing this operation with this account type combination is not allowed
AUTHORIZATION_INVALID_ACCOUNTHOLDER_HIERARCHY	Setting an account holder as its own parent or a creating a cycle in the hierarchy is not allowed
AUTHORIZATION_INVALID_DEBIT	Authorization failed
AUTHORIZATION_MAX_TRANSFER_ACCOUNT_THROUGHPUT	Maximum transfer account throughput amount including fee threshold exceeded
AUTHORIZATION_MAX_TRANSFER_ACCOUNT_THROUGHPUT_RECEIVER	Maximum transfer account throughput amount including fee threshold exceeded for receiver
AUTHORIZATION_MAX_TRANSFER_ACCOUNT_THROUGHPUT_SENDER	Maximum transfer account throughput amount including fee threshold exceeded for sender
AUTHORIZATION_MAX_TRANSFER_AMOUNT	Maximum transfer amount threshold exceeded
AUTHORIZATION_MAX_TRANSFER_AMOUNT_FEE	Maximum transfer amount including fee threshold exceeded



AUTHORIZATION_MAX_TRANSFER_AMOUNT_FEE_RECEIVER	Maximum transfer amount including fee threshold exceeded for receiver
AUTHORIZATION_MAX_TRANSFER_AMOUNT_FEE_SENDER	Maximum transfer amount including fee threshold exceeded for sender
AUTHORIZATION_MAX_TRANSFER_AMOUNT_RECEIVER	Maximum transfer amount threshold exceeded for receiver
AUTHORIZATION_MAX_TRANSFER_AMOUNT_SENDER	Maximum transfer amount threshold exceeded for sender
AUTHORIZATION_MAX_TRANSFER_TIMES	Maximum transfer counter threshold exceeded
AUTHORIZATION_MAXIMUM_AMOUNT_ALLOWED_TO_RECEIVE	Maximum receive amount threshold exceeded
AUTHORIZATION_MAXIMUM_AMOUNT_ALLOWED_TO_SEND	Maximum send amount threshold exceeded
AUTHORIZATION_MAXIMUM_AMOUNT_TO_APPROVE	Authorization failed. Maximum threshold limit reached for Approver
AUTHORIZATION_MINIMUM_AMOUNT_ALLOWED_TO_SEND	Amount below minimum amount threshold
AUTHORIZATION_MINIMUM_AMOUNT_ALLOWED_TO_SEND_ACCOUNT	Amount below minimum amount threshold for account
AUTHORIZATION_MINIMUM_AMOUNT_ALLOWED_TO_SEND_ACCOUNT_REFERENCE	Amount below minimum amount threshold for account reference
AUTHORIZATION_NOT_REGISTERED_STATUS	Account holder is not registered
AUTHORIZATION_RECEIVER_ACCOUNT_NO_DEPOSIT	Receivers account does not allow deposit
AUTHORIZATION_RECEIVER_FIRST_DEPOSIT_MINIMUM_AMOUNT_ACCOUNT	Could not perform operation because amount is below minimum first deposit amount threshold
AUTHORIZATION_RECEIVER_MAX_ALLOWED_BALANCE	Maximum balance threshold for receiver exceeded
AUTHORIZATION_RECEIVING_ACCOUNT_NOT_ACTIVE	Receivers account is not active
AUTHORIZATION_RECEIVING_ACCOUNT_UNAVAILABLE	Receiving account is unavailable
AUTHORIZATION_SENDER_ACCOUNT_NO_WITHDRAWAL	Withdrawal from recipient account not allowed
AUTHORIZATION_SENDER_ACCOUNT_NOT_ACTIVE	Senders account is not active
AUTHORIZATION_SENDER_MIN_ALLOWED_BALANCE	Minimum balance threshold for sender exceeded
AUTHORIZATION_SENDING_ACCOUNT_UNAVAILABLE	Sending account is unavailable
AUTO_REFILL_ALREADY_EXISTS_FOR_RESOURCE	Auto refill for given resource already exists
AUTO_REFILL_DOES_NOT_EXIST_FOR_RESOURCE	Auto refill does not exist for given resource
BANK_ACCOUNT_NOT_REGISTERED	A bank account with given bank code and account number could not be found
BANK_ACCOUNT_ROUTE_NOT_FOUND	A route to the wanted bank account could not be found
BANK_TRANSACTION_ID_NOT_ALLOWED	BankTransactionId is not allowed if command is of type ROLLBACK
BANK_TRANSACTION_ID_REQUIRED	BankTransactionId is required if command is of type COMPLETE
BANKDOMAIN_MISSING	No bank domain set in the system.
BANKDOMAIN_NOT_FOUND	Bank domain with given ID was not found



BANKDOMAIN_USED	Could not perform operation since the Bank domain is currently in use
BANKDOMAIN_WITH_NAME_ALREADY_EXISTS	Could not perform operation since Bank domain with given name already exists
BLOCK_ACCOUNT_FAILED	Account could not be blocked
BULK_QUOTE_ID_NOT_FOUND	Provided Bulk Quote ID was not found in the server.
BULK_TRANSFER_ID_NOT_FOUND	Provided Bulk Transfer ID was not found in the server.
CAN_NOT_RECEIVE_CASHIN	Receiving account holder has insufficient permission to receive cashin
CAN_NOT_RECEIVE_CASHOUT	Receiving account holder has insufficient permission to receive cashout
CANNOT_INVITE_ADMIN	A group admin cannot be invited to a group.
CANNOT_REMOVE_ADMIN	Group administrator cannot be removed from the group.
CANNOT_SWITCH_PAYOUT_CYCLE_WITH_SELF	Contributor cannot change payout cycle with self.
CERTIFICATE_ALREADY_UPDATED	Certificate already updated for the user
CERTIFICATE_CREDENTIALS_ALREADY_EXIST	Certificate credentials already exist
CERTIFICATE_CREDENTIALS_NOT_FOUND	Certificate credentials do not exist
CLOSE_ACCOUNT_FAILED	Account could not be closed
COMMUNICATION_ERROR	A communication error occurred.
COMPANY_INFORMATION_ALREADY_EXIST	The company information already exist
COMPANY_INFORMATION_DOES_NOT_EXIST	The company information does not exist
CONTACT_ALREADY_EXISTS	Contact was not added since it already exists
CONTACT_NOT_FOUND	Contact with given FRI could not be found
COULD_NOT_CREATE_ACCOUNTHOLDER_CREDENTIALS	Could not create credentials for account holder
COULD_NOT_CREATE_ACCOUNTHOLDER_SETTINGS	Could not create account holder
COULD_NOT_PERFORM_ADD_ACCOUNT_OPERATION	Could not perform add account operation
COULD_NOT_PERFORM_APPROVAL_OF_OPERATION_PLEASE_RETRY_OPERATION_FROM_BEGINNING	Failed to approve the operation, please retry the initial operation again
COULD_NOT_PERFORM_ATM_PARTIAL_REVERSAL_OPERATION	ATM partial reversal operation could not be performed
COULD_NOT_PERFORM_ATM_REDEEM_VOUCHER_OPERATION	ATM redeem voucher operation could not be performed
COULD_NOT_PERFORM_ATM_ROLLBACK_VOUCHER_OPERATION	ATM rollback voucher operation could not be performed
COULD_NOT_PERFORM_OPERATION	Could not perform the operation. Additional reason attached.
COULD_NOT_PERFORM_OPERATION_ACCOUNT_NOT_LINKED	Could not perform the operation, the account has to be linked with sp account.
COULD_NOT_PERFORM_TRANSACTION	Transaction could be performed



COULD_NOT_RECEIVE_REMITTANCE	Receive remittance failed
COULD_NOT_UPDATE_ACCOUNTHOLDER_SETTINGS	Could not perform update of account holder settings
COULD_NOT_UPDATE_PENDING_PAYMENT	Could not perform update of pending payment status
COULD_NOT_UPDATE_PENDING_WITHDRAW	Could not perform update of pending withdraw status
COUNTER_CONFIGURATION_NOT_FOUND	The specified counter configuration does not exist.
CREDENTIAL_IS_NOT_SUSPENDED	The credential is not in the expected suspended state
CREDENTIAL_NOT_ACTIVE	Operation not allowed because credential not in active state
CREDENTIAL_REPEATED_SECRET_DOES_NOT_MATCH	Repeated secrets does not match each other
CREDENTIAL_TYPE_NOT_FOUND	Given credential type does not exist in the system
CREDENTIALS_NOT_FOUND	The user does not have any credential of the given type
CREDENTIALS_NOT_FOUND	Given credentials could not be found
CURRENCIES_DONT_MATCH	For this operation currency conversion is not allowed
CURRENCY_NOT_SUPPORTED	Given currency is not supported in the system
CUSTODY_ACCOUNT_ALREADY_EXISTS	Custody Account already exists
CUSTODY_ACCOUNT_CERTIFICATE_ALREADY_EXIST	Custody account certificate already exist
CUSTODY_ACCOUNT_CERTIFICATE_COULD_NOT_BE_DELETED	Custody account certificate could not deleted
CUSTODY_ACCOUNT_CERTIFICATE_COULD_NOT_BE_EMPTY	Custody account certificate could not be empty
CUSTODY_ACCOUNT_CERTIFICATE_COULD_NOT_BE_USED_FOR_NON_REPUDIATION	Custody account certificate could not be used for non-repudiation
CUSTODY_ACCOUNT_CERTIFICATE_DOES_NOT_EXIST	Custody account certificate does not exist
CUSTODY_ACCOUNT_CERTIFICATE_ERROR	Custody account certificate error
CUSTODY_ACCOUNT_CERTIFICATE_NO_SUCH_ALGORITHM	Unsupported algorithm for custody account certificate
CUSTODY_ACCOUNT_CERTIFICATE_WITH_SAME_ID_ALREADY_EXIST	Custody account certificate with same id already exist
CUSTODY_ACCOUNT_NAME_AND_BANKDOMAIN_SAME_AS_BEFORE	Name and Bank Domain are same as before
CUSTODY_ACCOUNT_NOT_FOUND	Could not find CustodyAccount since Custody Account with provided account number and bank code does not exists
CUSTODY_ACCOUNT_STILL_CONTAINS_MONEY	Custody account still contains money
CUSTODY_ACCOUNT_STILL_HAVE_PENDING_TRANSACTION	This Custody account still have pending transaction
CUSTODY_ACCOUNT_UPDATION_FAILED_NAME_AND_BANKDOMAIN_MISSING	Could not update custody account as name and bank domain both are missing
DATEFORMAT_INVALID	DateTime string could not be parsed
DESTINATION_FSP_ERROR	Destination FSP does not exist or cannot be found.



DIFFERENT_BANKDOMAINS	Account holder and account must belong to the same bank domain
DOCUMENT_DOES_NOT_EXIST	The document does not exist
DOCUMENT_URL_CONFIGURATION_ERROR	The document rules has been incorrectly configured in the system
DOCUMENT_URL_INVALID	This document URL is invalid
DOCUMENT_URL_NOT_ALLOWED	This document URL is not allowed
DUPLICATE_NOTIFICATION_GROUP_FOUND	Duplicate Notification group found
DUPLICATED_COMMISSION_OPERATION_TYPE	Duplicated commissioned operation type found
DUPLICATED_CREDENTIAL_TYPES	Duplicated credential types given when creating credentials for account holder
END_DATE_IN_FUTURE	End date can not occur in the future
END_USER_SERVICE_DENIED	Service was denied for the user by an external system
ERROR_RESPONSE	An callback error response was sent from a client.
EXPIRED_OR_INVALID_TRANSACTION_ID	Transaction ID is either invalid or this transaction has expired
EXTERNAL_ACCOUNT_HOLDER_ERROR	Unknown error related to external account holder.
EXTERNAL_ACCOUNT_HOLDER_INACTIVE	External account holder is inactive.
EXTERNAL_ACCOUNT_HOLDER_LIMIT_ERROR	External account holder is not allowed to be involved in the requested operation.
EXTERNAL_ACCOUNT_HOLDER_NOT_FOUND	External account holder could not be found.
EXTERNAL_ACCOUNT_HOLDER_REJECTED	External account holder or their provider rejected request.
EXTERNAL_CLIENT_ERROR	Generic client error.
EXTERNAL_COMMUNICATION_ERROR	Failure to communicate with external system.
EXTERNAL_CURRENCY_UNSUPPORTED	External system does not support the requested currency.
EXTERNAL_ERROR_MESSAGE	This error originates from an external system
EXTERNAL_FINANCIAL_SERVICE_PROVIDER_NOT_FOUND	External financial service provider could not be found.
EXTERNAL_OPERATION_UNSUPPORTED	External server does not support the requested transaction type.
EXTERNAL_PAYMENT_INSTRUMENT_ALREADY_EXISTS	External Payment Instrument with given identifier was not added since it already exists
EXTERNAL_PAYMENT_INSTRUMENT_CONFIGURATION_MISMATCH	Wrong External Payment Instrument Register called for this configuration
EXTERNAL_PAYMENT_INSTRUMENT_IN_ILLEGAL_STATE	External Payment Instrument in illegal state
EXTERNAL_PAYMENT_INSTRUMENT_NOT_FOUND	External Payment Instrument with given identifier could not be found
EXTERNAL_PAYMENT_INSTRUMENT_PROCESS_ERROR	Error processing an External Payment Instrument
EXTERNAL_PAYMENT_INSTRUMENT_REMOVE_FAILED	External Payment Instrument could not be removed



EXTERNAL_PAYMENT_INSTRUMENT_VALIDATION_FAILED	External Payment Instrument validation failed
EXTERNAL_REJECTED	External system rejected the transfer.
EXTERNAL_RESOURCE_EXPIRED	External resource has expired.
EXTERNAL_RESOURCE_NOT_FOUND	External server could not find resource.
EXTERNAL_SERVER_ERROR	External server could not handle request.
EXTERNAL_SERVICE_PROVIDER_NOT_FOUND	External service provider with given User name could not be found
EXTERNAL_VALIDATION_FAILED	External server failed to validate request.
EXTERNAL_VERSION_ERROR	External server does not supported the requested version.
FAILED_TO_CREATE_NETTING_REPORT	Netting report error
FAILED_TO_DELETE_CUSTODY_ACCOUNT	Failed to delete custody account.
FAILED_TO_PERFORM_EXTERNAL_OPERATION	Failed to perform operation towards external system.
FEE_REVERSAL_NOT_ALLOWED	Fees reversal is not allowed after certain time has elapsed
FIELD_TOO_LONG	Data in element is too long
FINANCIAL_CONFIGURATION_DOES_NOT_EXIST	Financial Configuration does not exist
FINANCIAL_CONTROLLER_TRANSACTION_NOT_FOUND	Neither transaction with transaction ID nor original transaction with original transaction ID could be found
FINANCIAL_RESOURCE_NOT_SUPPORTED	Financial resource not supported
FINANCIAL_TRANSACTION_RECEIPT_FILE_CREATION_FAILED	Unable to create receipt file for financial transaction
FIRST_NAME_NOT_ALLOWED_IN_PASSWORD	Using first name in password is not allowed
FLOAT_TRANSFER_BETWEEN_HIERARCHIES_EXEMPTION_DOES_NOT_EXIST	Float transfer between hierarchies rule to be removed does not exist
FLOAT_TRANSFER_BETWEEN_HIERARCHIES_INVALID_RULE	It is not possible to set an exemption for an agent with an invalid/disallowed rule
FLOAT_TRANSFER_BETWEEN_HIERARCHIES_NULL_PARENT	It is not possible to remove an exemption for a sender agent who's parent is 'null'
FLOAT_TRANSFER_EXEMPTION_ALREADY_EXISTS	The specified float transfer exemption rule already exists
FRI_CAN_ONLY_HAVE_ONE_OWNER	Setting multiple owners of an FRI is not allowed
FRI_INVALID	FRI string is incorrectly formatted
FRI_REFERENCE_NOT_FOUND	
FROM_DATE_AFTER_TO_DATE	Operation requires that start date has to occur before end date
FUNDS_STILL_LEFT	Could not perform operation as funds are still left
FUTURE_FROM_DATE_AND_FUTURE_TO_DATE	Operation requires that start date and end date should not be in future
GAP_EXISTS_IN_PAYOUT_CYCLE_ORDER	A payout cycle has not been assigned to a contributor.



GENERIC_SERVER_ERROR	Generic server error.
GROUP_ACCOUNT_CREATION_FAILED	Group account could not be created.
GROUP_ALREADY_USED_BY_ANOTHER_BANKDOMAIN	Could not perform operation since the Group is already used by another Bank domain
GROUP_CONTAINS_NONACCEPTED_CONTRIBUTORS	Group contains contributors who have not accepted their invitations.
GROUP_CONTAINS_NONACCEPTED_MEMBERS	Group contains members who have not accepted their invitations.
GROUP_CYCLE_NOT_READY_FOR_MANUAL_PAYOUT	The given payout cycle has not started and is not ready for manual payout.
GROUP_MEMBER_COULD_NOT_BE_REMOVED	Group member could not be removed from group
GROUP_MEMBER_COULD_NOT_BE_REMOVED_FROM_BANKDOMAIN	Group member could not be removed from Bank domain
GROUP_MEMBER_NOT_FOUND	Group member with given ID could not be found in this group
GROUP_NOT_FOUND	Group could not be found.
GROUP_USED_BY_BANKDOMAIN	Group used by Bank domain is not allowed to be removed
HIGH_CONTENTION_ACCOUNT_NOT_SUPPORTED_AS_MIRRORED_ACCOUNT	It's not possible to mirror a High Contention account in SDP since it doesn't have a balance.
HOME_CHARGING_REGION_NOT_FOUND	Home charging region with given ID was not found
HOME_CHARGING_REGION_USED	Could not perform operation since the home charging region is currently in use
HOME_CHARGING_REGION_WITH_NAME_ALREADY_EXISTS	Could not perform operation since Home Charging Region with given name already exists
ID_LENGTH_OUT_OF_BOUNDS	Length of identification is not within length boundaries
IDENTIFICATION_ID_AND_TYPE_ALREADY_EXIST	Duplicate identification
IDENTITY_INVALID	Identity string is incorrectly formatted
IDTYPE_NOT_SUPPORT	ID type not supported by ISO 20022
ILLEGAL_CREDENTIAL_STATUS_CHANGE	The credential status change is illegal
ILLEGAL_GROUP_ROLE	The account holder cannot have the provided role.
ILLEGAL_GROUP_STATE	The group cannot be changed to the given state.
IMSI_INVALID	Format of IMSI is invalid
INACTIVE_ACCOUNT	Inactive receiver account
INCORRECT_PASSWORD	Incorrect password
INCORRECT_PIN	Incorrect pin code
INCORRECT_SECURITY_ANSWER	Security answer was not correct
INCORRECT_TRANSACTION_STATUS	Transaction state is incorrect for this operation
INITIATOR_ALREADY_HAS_PENDING_SWITCH_PAYOUT_CYCLE	The initiator of switching a payout cycle with another contributor already has a pending switch.
INITIATOR_IS_NOT_ACCEPTED_CONTRIBUTOR_IN_THE_GROUP	Initiator has not accepted the invitation to group.



INTERNAL_ERROR	An internal error caused the operation to fail
INTERNAL_LOAN_ACCOUNT_NOT_FOUND	Could not find Internal Loan Account with given Id
INTERNAL_LOAN_AMOUNT_MIN_LARGER_THAN_AMOUNT_MAX	Amount Min cannot be larger than Amount Max
INTERNAL_LOAN_APPLICATION_IS_NOT_VALID	The state of this Internal Loan Application is not valid for this operation
INTERNAL_LOAN_APPLICATION_NOT_FOUND	Could not find Internal Loan Application
INTERNAL_LOAN_MAXIMUM_NUMBER_OF_APPLICATIONS	Maximum number of Internal Loan Applications reached
INTERNAL_LOAN_PRODUCT_ALREADY_EXIST	Internal Loan Product already exist
INTERNAL_LOAN_PRODUCT_IS_NOT_ACTIVE	The Internal Loan Product status is not active
INTERNAL_LOAN_PRODUCT_NOT_ALLOWED	The given Internal Loan Product is not allowed to be used
INTERNAL_LOAN_PRODUCT_NOT_FOUND	Could not find Internal Loan Product with given identity and product code
INTERNAL_LOAN_PROVIDER_ALREADY_EXIST	Internal Loan Provider already exist
INTERNAL_LOAN_PROVIDER_NOT_FOUND	Could not find Internal Loan Provider with given identity
INVALID_ACCOUNT_FRI	The FRI must be of account type
INVALID_ACCOUNT_HOLDER_ADDITIONAL_INFORMATION_NAME	The specified account holder's additional information name is not valid or does not exist
INVALID_ACCOUNT_TRANSACTION_STATISTICS_RESOLUTION	The given account transaction statistics resolution type is invalid
INVALID_ACCOUNTHOLDER_SETTINGS	Account holder settings are invalid
INVALID_APPROVAL_TRANSACTION_STATUS	Approval transaction has invalid state
INVALID_ATM_OPERATION	Invalid ATM operation
INVALID_CERTIFICATE	Certificate is invalid
INVALID_CHOSEN_DELIVERY_METHOD	Invalid Delivery method
INVALID_COMMISSION_OPERATION_TYPE_NAME	Operation is not commissioning-enabled, or doesn't have entry in OperationType
INVALID_COUNTER_IDENTIFIER	Profile counter identifier validation
INVALID_CREDENTIAL_TYPE	The given credential type is not supported when creating credentials for account holder
INVALID_CURRENCY	Currency is either not valid or used incorrectly
INVALID_FIELD_VALUE	Element has invalid format
INVALID_FINANCIAL_TRANSACTION_ROLE	An invalid financial transaction role is sent to M3
INVALID_IDENTITY_LISTING_NOTIFICATION_GROUP	Invalid user identity
INVALID_NOTIFICATION_CHANNEL	The notification channel is invalid or cannot be used
INVALID_PROFILE	Profile validation
INVALID_PROFILE_COUNTER	Profile counter validation
INVALID_PROFILE_NAME	Profile name validation



INVALID_PROFILE_THRESHOLD	Profile threshold validation
INVALID_RECEIVER	Invalid receiving end user
INVALID_REMITTANCE_STATUS	Incorrect remittance status
INVALID_SCHEDULED_TRANSACTION_DATE	A transaction can only be scheduled from a future date
INVALID_SCHEDULED_TRANSACTION_STATUS	Invalid scheduled transaction status for the requested operation
INVALID_SETTLEMENT_DEPOSIT_ID	Settlement deposit with given ID could not be found
INVALID_SETTLEMENTID	Settlement with given ID could not be found
INVALID_SIGNATURE	The provided signature is not correct
INVALID_STATUS	Current status is not as expected
INVALID_TRANSACTION_TYPE	Invalid Transaction type
INVALID_USER_APPROVAL_TYPE	Invalid User (or financial) approval type or not registered
INVOICE_ALREADY_IN_FINALIZED_STATE	The invoice is already in a finalized state.
INVOICE_HAS_EXPIRED	The invoice has expired.
INVOICE_NOT_FOUND	The invoice was not found.
INVOICE_SENDER_FRI_IS_SAME_AS_RECEIVER_FRI	The sender's FRI is same as receiver's FRI.
LAST_BANKDOMAIN_NOT_ALLOWED_TO_REMOVE	Bank domain could not be removed since it is the last remaining
LIMITED_RETRIES_FOR_SERVICE_PROVIDER_NOT_ALLOWED	Limited retries are not allowed for the specified service provider
LOAN_APPLICATION_NOT_FOUND	Could not find Loan Application with given approval ID
LOAN_PROVIDER_NOT_FOUND	Could not find Loan Provider with given name
MANDATORY_FIELD_MISSING	Mandatory element is missing
MAX_ACCOUNTS_LIMIT_REACHED	Reached limit for max accounts
MAX_GROUP_CREATIONS_IN_PERIOD_REACHED	Maximum number of groups created for a given period has been reached.
MAX_GROUP_INVITATIONS_SENT_IN_PERIOD_REACHED	Maximum number of group invitations for a given period has been reached.
MAX_NUMBER_OF_CONTACTS_EXCEEDED	Maximum number of contacts threshold exceeded
MAX_NUMBER_OF_DOCUMENT_EXCEEDED	Maximum number of documents exceeded
MAX_NUMBER_OF_MEMBERSHIPS_REACHED	Maximum number of group memberships has been reached.
MAX_ONE_ADMIN_PER_GROUP	Only one administrator is allowed per group.
MAX_PREAPPROVAL_VALIDITY_DURATION_EXCEEDED	The requested pre-approval validity duration exceeds the configured max value
MAXIMUM_INVOICE_VALIDITY_DURATION_EXCEEDED	The validity duration exceeds the system configured maximum.



MEMBER_ALREADY_ADDED_WITH_SAME_ROLE	The given group member has already been added to the group with the given role.
MEMBER_IS_NOT_ACTIVE	Group member is inactive.
MEMBER_LACKS_ACTIVE_ACCOUNT_WITH_CORRECT_CURRENCY	The given group member does not have an account in the currency of the group.
MEMBER_LACKS_PERMISSION	The given group member does not have permission for their role in the group.
MISSING_ACTOR_MSISDN_FOR_EXTERNAL_RATING	Actor MSISDN is missing
MISSING_EXCHANGE_RATE	Exchange rate for conversion between currencies does not exist.
MISSING_GROUP_STATUS	Group status is missing.
MISSING_MANDATORY_ELEMENT	Mandatory element in the data model was missing.
MISSING_MEMBER_ID	Group member id is missing.
MODIFIED_REQUEST	Request with the same ID has previously been processed in which the parameters are different.
MORE_THAN_ONE_MUTUALLY_EXCLUSIVE_OPTIONAL_PARAMETERS_SET	More than one of a set of optional but mutual parameters were set.
MSISDN_ALREADY_USED_BY_POINTOFSALE	Given msisdn already used by a point of sale
NO_ACCESS	User has no access permission
NO_CONTRIBUTOR_WITH_SPECIFIED_PAYOUT_CYCLE	There is no contributor for the provided payout cycle number.
NO_CURRENTDAY_STATISTICS	No current day statistics available to export
NONASCII_CHARACTERS_IN_PASSWORD	Using non ASCII characters in password is not allowed
NONDIGIT_CHARACTERS_IN_PINCODE	Non-digit characters in pin code
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN	It is not allowed to change bank domain for an account holder
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_ACCOUNT_HAS_ACTIVE_RESERVATIONS	It is not allowed to change bank domain since the account holder has active reservations
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_ALREADY_ASSIGNED_TO_BANK_DOMAIN	It is not allowed to change bank domain since the account holder is already assigned to the specified bank domain
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_CONNECTED_TO_OTHER_ACCOUNT	It is not allowed to change bank domain since the account holder is connected to account owned by other account holder
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_HAS_ACCOUNT_WITH_NEGATIVE_BALANCE	It is not allowed to change bank domain since the account holder has an account with negative balance
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_HAS_EXTERNAL_ACCOUNT	It is not allowed to change bank domain since the account holder has external account
NOT_ALLOWED_TO_CHANGE_BANKDOMAIN_OTHER_ACCOUNT_HOLDERS_CONNECTED	It is not allowed to change bank domain since other account holders are connected to the account holder's account
NOT_ALLOWED_TO_VIEW_INVOICE	User doesn't have permission to view specified invoice



NOT_AUTHORIZED	The logged in user does not have sufficient permissions to perform this operation
NOT_ENOUGH_FUNDS	Not enough funds or loyalty points to perform operation
NOT_IMPLEMENTED	Service requested is unsupported by the server.
NOTIFICATION_GROUP_ACCOUNT_HOLDER_NOT_FOUND	Notification group supervisor not found
NOTIFICATION_GROUP_DUPLICATE_ACCOUNT_HOLDER_FOUND	Duplicate Notification group account holder found
NOTIFICATION_GROUP_DUPLICATE_SUPERVISOR_FOUND	Duplicate Notification group supervisor found
NOTIFICATION_GROUP_NO_PERMISSION_TO_BECOME_SUPERVISOR	Permission Not available tp become supervisor
NOTIFICATION_GROUP_NOT_FOUND	Invalid notification group identity
NOTIFICATION_GROUP_SUPERVISOR_ADMIN_NOT_FOUND	Admin supervisor not found
NOTIFICATION_GROUP_SUPERVISOR_NOT_FOUND	Notification group supervisor not found
ONE_CUSTODY_ACCOUNT_MUST_EXIST	One custody account must exist
ONLY_ONE_REFUND_PER_DEBIT_IS_ALLOWED	Only one refund per debit is allowed
OPERATION_NOT_SUPPORTED_ON_EXTERNAL_ACCOUNT	Operation not supported on external account
OPERATION_NOT_SUPPORTED_ON_MIGRATED_ACCOUNT	Operation not supported on a migrated account
OTHER_FINANCIAL_ERROR	Other financial error
OTP_FOR_BANK_ACCOUNT_VERIFICATION_MISSING	Request did not contain any OTP
OTP_FOR_EMAIL_VERIFICATION_INCORRECT	Request contained incorrect OTP
OTP_FOR_EMAIL_VERIFICATION_MISSING	Request did not contain any OTP
OTP_FOR_MSISDN_VERIFICATION_INCORRECT	Request contained incorrect OTP
OTP_FOR_MSISDN_VERIFICATION_MISSING	Request did not contain any OTP
OTP_INVALID	Format of OTP is invalid
OTP_NOT_FOUND	OTP not found, has expired or is used for the wrong operation
PASSWORD_ALREADY_USED_BEFORE	Password has already been used before
PASSWORD_TOO_SHORT	Password too short
PAYEE_FSP_UNSUPPORTED_TRANSACTION_TYPE	Transaction type unsupported or rejected by Payee FSP.
PAYEE_UNSUPPORTED_CURRENCY	Payee does not have an account which supports the requested currency.
PAYER_FSP_UNSUPPORTED_TRANSACTION_TYPE	Transaction type unsupported or rejected by Payer FSP.
PAYOUT_ALREADY_COMPLETED_FOR_PAYOUT_CYCLE	Payout has already been completed for the given payout cycle.
PAYOUT_ALREADY_INITIATED_FOR_APPROVER	The approver of switching a payout cycle with another contributor has already received a payout.
PAYOUT_ALREADY_INITIATED_FOR_INITIATOR	The initiator of switching a payout cycle with another contributor has already received a payout.
PAYOUT_COULD_NOT_BE_MADE_TO_CONTRIBUTOR	A payout could not be performed to the given contributor.



PAYOUT_CYCLE_ALREADY_ASSIGNED	The given payout cycle number has already been assigned to another contributor.
PAYOUT_CYCLE_NOT_WITHIN_BOUNDS	The given payout cycle number must be within the number of contributors in the group.
PAYOUT_CYCLE_ONLY_ALLOWED_FOR_CONTRIBUTOR	A payout cycle number is only allowed for contributors.
PAYOUT_CYCLE_REQUIRED_FOR_CONTRIBUTOR	A contributor must have a payout cycle number.
PENDING_COMMISSION_NOT_FOUND	Could not find Pending Commission
PICKUP_REFERENCE_ID_NOT_FOUND	Pickup with given MTCN could not be found
PINCODE_ALREADY_USED_BEFORE	PIN code has already been used
PINCODE_DENIED_BY_MATCHING_BIRTH_DATE	PIN code denied by matching birth date
PINCODE_DENIED_BY_MATCHING_IDENTIFICATION_NUMBER	PIN code denied by matching identification number
PINCODE_DENIED_BY_MATCHING_MSISDN	PIN code denied by matching MSISDN
PINCODE_DENIED_BY_MATCHING_REGULAR_EXPRESSION	PIN code denied by matching regular expression
PINCODE_HAS_EXPIRED	Wrong profile
PINCODE_TOO_LONG	Pin code too long
PINCODE_TOO_SHORT	Pin code too short
POINTOFSALE_NOT_ACQUIRED	The point of sale with given msisdn is not acquired.
PREAPPROVAL_NOT_FOUND	The pre-approval with given ID could not be found
PREAPPROVALS_NOT_FOUND	No pre-approvals found
PROFILE_ALREADY_EXISTS	Profile name validation
PROFILE_JSON_INVALID_FILE	Json File validation
PROFILE_MISMATCH	A profile mismatch between for example a provided counter configuration (belonging to one profile) and provided profile reference (having a different profile)
PROFILE_NOT_FOUND	Profile not found
PROFILE_TYPE_MISMATCH	Profile type validation
PROFILE_VERSION_MANDATORY	Profile version validation
PROFILE_VERSION_MISMATCH	Profile version validation
PROFILE_WITH_NAME_DOES_NOT_EXIST	Tag was not found
PROVIDER_CATEGORY_LINKED_WITH_ACCOUNT_HOLDER	Provider Category can not be deleted as it is linked with account holder
PROVIDER_CATEGORY_NOT_FOUND	No provider category with this name was found in the system
QUEUED_FOR_APPROVAL	Request is queued for approval
QUOTE_ALREADY_USED	Quote has already been used
QUOTE_EXPIRED	Quote has expired
QUOTE_EXPIRED	Client requested to use a quote which has already expired.



QUOTE_ID_NOT_FOUND	Provided Quote ID was not found in the server.
QUOTE_NOT_FOUND	Quote not found
QUOTE_NOT_SUPPORTED	Quote is not supported with the combination of parameters
QUOTE_PROCESSING	Quote is already processing
REACTIVATE_ACCOUNT_FAILED	Account could not be re-activated
REASON_NOT_ALLOWED	Reason is not allowed if command is of type COMPLETE
REASON_REQUIRED	Reason is required if command is of type ROLLBACK
RECONCILIATION_PROCESS_ERROR	Failed to process reconciliation file.
RECURRING_SCHEDULED_TRANSACTION_EXCEEDS_MAX_DAYS	Recurring scheduled transaction can only be scheduled a limited number of days into the future
REFERENCE_ID_ALREADY_IN_USE	This client reference ID is already used
REFERENCE_ID_NOT_FOUND	A client reference with this ID could not be found
REFUND_AMOUNT_CANNOT_EXCEED_THE_DEBIT_AMOUNT	Refund amount cannot be larger than the debit amount
RELATIVE_DOES_NOT_EXIST	The relative does not exist
REMITTANCE_ID_DOES_NOT_EXIST	Remittance Id does not exist in dbase
REMITTANCE_NOT_FOUND	Remittance not found, Invalid provider name or remittance id
REMITTANCE_NOT_POSSIBLE	Remittance not possible because given data is invalid.
REMOVE_PARENT_NOT_SUPPORTED_FOR_NONLEAF_CHILD	Removing a parent for agent who has children in the hierarchy is not allowed with child permissions.
REQUIRED_FIELD_MISSING	Element needed for this invocation is missing
REQUIRED_PASSWORD_CRITERIA_NOT_MATCH	Password criteria configuration invalid
RESOURCE_NOT_ACTIVE	Required resource is not active
RESOURCE_NOT_AVAILABLE	FRI not available
RESOURCE_NOT_FOUND	FRI not found
RESOURCE_TEMPORARY_LOCKED	Required resource is temporarily locked
REVERSAL_ALREADY_PERFORMED_ON_THIS_TRANSACTION	Reversal has already been performed on this transaction
REVERSAL_INITIALIZATION_FAILED	Reversal initialization failed
REVERSAL_NOT_ALLOWED	Reversal is not allowed after certain time has elapsed
REVERSAL_NOT_ALLOWED_ON_THIS_TRANSACTION_TYPE	Reversal is not allowed on this type of transaction
REVERSAL_SUB_TRANSACTION_INVALID	Reversal not possible since sub transaction is invalid
REVERSED_USER_ID_NOT_ALLOWED_IN_PASSWORD	Using reversed user id in password is not allowed
ROLLBACK_NOT_POSSIBLE	Transaction could not be performed since neither of the parties supports rollback



SCHEDULED_TRANSACTION_EXCEEDS_MAX_DAYS	Transaction can only be scheduled a limited number of days into the future
SCHEDULED_TRANSACTION_NOT_FOUND	The requested scheduled transaction could not be found
SECRETARY_ALREADY_ASSIGNED	Only one secretary is allowed per group.
SELF_SERVICE_NOTIFICATION_DUPLICATE	This self service notification already exists.
SELF_SERVICE_NOTIFICATION_LIMIT_EXCEEDS_HARD_LIMIT	The notification limit exceeds the hard limit
SELF_SERVICE_NOTIFICATION_NOT_FOUND	
SELF_SERVICE_NOTIFICATION_TOO_MANY	Max number of self serviced notifications reached, cannot create another.
SERVICE_PROVIDER_DOES_NOT_EXIST	Service Provider does not exist
SERVICE_PROVIDER_GATEWAY_URL_ALREADY_EXIST	Service Provider already has gateway URL defined
SERVICE_PROVIDER_TYPE_NOT_FOUND	Service provider type not found
SERVICE_TEMPORARILY_UNAVAILABLE	The operation could not be performed right now but might succeed later
SERVICE_UNAVAILABLE	Service Unavailable, a resource is not accessible at the moment, try again later.
SESSION_INVALID	The session has timed out
SETTLEMENT_AMOUNT_DO_NOT_MATCH	The response amount does not match with the amount in the Payment Instruction
SETTLEMENT_ANOTHER_SETTLEMENT_BEING_PROCESSED	Another settlement task is ongoing.
SETTLEMENT_BANK_INSTRUCTION_IN_WRONG_STATE	Settlement bank instruction in wrong state
SETTLEMENT_BANK_INSTRUCTION_NOT_FOUND	Settlement bank instruction could not be found
SETTLEMENT_CLOSING_BALANCE_MISMATCH	Balance reconciliation failed. Provided closing balance does not match with the expected closing balance of custody account
SETTLEMENT_CONFIGURATION_CERTIFICATE_DOES_NOT_EXIST_FOR_CUSTODY_ACCOUNT	Certificate does not exist for custody account not able to verify signature of settlement file
SETTLEMENT_CONFIGURATION_CUSTODY_ACCOUNT_ALREADY_EXISTS_WITH_SETTLEMENT_CONFIGURATION	Custody account already exists with settlement configuration
SETTLEMENT_CONFIGURATION_DOES_NOT_EXIST_FOR_CUSTODY_ACCOUNT	Could not find settlement Configuration for custody account
SETTLEMENT_CONFIGURATION_DOES_NOT_SUPPORT_TRANSFERFROMBANK_OPERATION	Settlement Configuration does not support transfer from bank operation
SETTLEMENT_CONFIGURATION_FOR_IDENTIFIER_ALREADY_EXISTS	Settlement configuration for identifier already exists
SETTLEMENT_CONFIGURATION_INSTRUCTION_FORMATTER_CLASS_NOT_FOUND	Instruction formatter class not found
SETTLEMENT_CONFIGURATION_NOT_FOUND	Settlement configuration not found
SETTLEMENT_CONFIGURATION_SETTLEMENT_PARSER_CLASS_NOT_FOUND	Settlement parser class not found
SETTLEMENT_CONFIGURATION_UNABLE_TO_REMOVE_CUSTODY_ACCOUNT_ID	Unable to remove custody account id from settlement configuration



SETTLEMENT_CURRENCY_DO_NOT_MATCH	Currency does not match with custody account's currency.
SETTLEMENT_CUSTODY_ACCOUNT_DO_NOT_MATCH	Settlement process failed because the custody accounts information do not match.
SETTLEMENT_DEPOSIT_ALREADY_REVERSED	The deposit has already been reversed.
SETTLEMENT_DEPOSIT_IN_INVALID_STATE	The deposit is in an invalid state.
SETTLEMENT_DEPOSIT_NOT_FOUND	The deposit was not found.
SETTLEMENT_DEPOSIT_TRANSACTION_DETAILS_MISSING	The transaction details of the deposit were not found.
SETTLEMENT_DUPLICATE_RECORD_FOUND	Duplicated deposit or reverse deposit record found.
SETTLEMENT_FILE_DIGEST_VERIFICATION_ERROR	Digest does not exist or digest verification failed for settlement file
SETTLEMENT_FILE_FORMAT_ERROR	Error in parsing settlement file
SETTLEMENT_FILE_SIGNATURE_VERIFICATION_ERROR	Signature does not exist or signature verification failed for settlement file
SETTLEMENT_IDENTIFIER_DO_NOT_MATCH	Payment Instruction Response caller identifier mismatch
SETTLEMENT_IDENTIFIER_NOT_FOUND	Settlement identifier not found
SETTLEMENT_INVALID_AMOUNT	Invalid or no amount in Payment Instruction Response.
SETTLEMENT_INVALID_RECEIVER	Settlement error due to invalid receiver. There is no such account holder or custody account exists.
SETTLEMENT_NOT_READY	Settlement is in wrong state
SETTLEMENT_ONLINE_SETTLEMENT_NOT_ENABLED_FOR_CUSTODY_ACCOUNT	Online settlement not enabled for custody account
SETTLEMENT_OPENING_BALANCE_MISMATCH	Opening Balance reconciliation failed
SETTLEMENT_PAYMENT_INSTRUCTION_IN_WRONG_STATE	Wrong state for Settlement PaymentInstruction.
SETTLEMENT_PAYMENT_INSTRUCTION_NOT_FOUND	Settlement PaymentInstruction not found.
SETTLEMENT_PROCESS_ERROR	Failed to process settlement file.
SETTLEMENT_RECEIVING_ACCOUNT_DO_NOT_MATCH	The receiver account does not match
SETTLEMENT_TRANSFER_IN_WRONG_STATE	Wrong state for Settlement CustodyAccountsTransfers.
SOURCE_AND_TARGET_ARE_THE_SAME	Source is the same as the target
SOURCE_NOT_FOUND	Source FRI not found
START_DATE_INVALID	Group start date is invalid.
SURNAME_NOT_ALLOWED_IN_PASSWORD	Using surname in password is not allowed
SUSPEND_ACCOUNT_FAILED	Not possible to put account in suspended state
TAG_WITH_NAME_ALREADY_EXISTS	Tag could not be created because a Tag with submitted name already exists
TAG_WITH_NAME_DO_NOT_EXISTS	Tag was not found



TARGET_AUTHORIZATION_ERROR	Target financial resource has reached counter or balance limits, has insufficient funds or is missing permissions
TARGET_NOT_FOUND	Target FRI not found
TAX_REPORT_NOT_FOUND	Tax report not found
THRESHOLD_CONFIGURATION_NOT_FOUND	The specified threshold configuration does not exist.
TOO_FEW_CONTRIBUTORS	Group contains too few contributors.
TOO_FEW_LOWERCASE_ALPHABETIC_CHARACTERS_IN_PASSWORD	Too few lower case alphabetic characters in password
TOO_FEW_NUMERICAL_DIGITS_IN_PASSWORD	Too few digits in password
TOO_FEW_SPECIAL_CHARACTERS_IN_PASSWORD	Too few special characters in password
TOO_FEW_UPPERCASE_ALPHABETIC_CHARACTERS_IN_PASSWORD	Too few upper case alphabetic characters in password
TOO_FEW_UPPERCASE_OR_LOWERCASE_ALPHABETIC_CHARACTERS_IN_PASSWORD	Too few upper or lower case alphabetic characters in password
TOO_MANY_CONSECUTIVE_DIGITS_IN_PINCODE	Too many consecutive digits in PIN code
TOO_MANY_IDENTICAL_CONSECUTIVE_CHARACTERS_IN_PASSWORD	Too many identical consecutive characters in password
TOO_MANY_REPEATED_DIGITS_IN_PINCODE	Too many repeated digits in PIN code
TOTAL_BATCH_LIMIT_EXCEEDED	Total batch limit exceeded
TRANSACTION_ID_OR_ORIGINAL_TRANSACTION_ID_MISSING	Neither transaction ID nor original transaction ID was given
TRANSACTION_NOT_ALLOWED	Transaction was not allowed
TRANSACTION_NOT_COMPLETED	Transaction has not been completed and its state is either Pending or Failed
TRANSACTION_NOT_FOUND	Transaction not found
TRANSACTION_NOT_QUEUED	Transaction is not queued
TRANSACTION_REQUEST_APPROVED	Payer approved an already completed transaction request from a Payee.
TRANSACTION_REQUEST_EXPIRED	Client requested to use a transaction request which has already expired.
TRANSACTION_REQUEST_NOT_FOUND	Provided Transaction Request was not found in the server.
TRANSACTION_REQUEST_REJECTED	Payer rejected a transaction request from a Payee.
TRANSACTION_TIMED_OUT	Transaction timed out
TRANSFER_EXPIRED	Client requested to use a transfer which has already expired.
TRANSFER_ID_NOT_FOUND	Provided Transfer ID was not found in the server.
TRANSFER_TYPE_AND_ACCOUNT_DO_NOT_MATCH	Receiving account is invalid for this type of transaction
TRANSFERTOBANK_NOT_SUPPORTED	Operation is not supported for this account
TREASURER_ALREADY_ASSIGNED	Only one treasurer is allowed per group.



UNABLE_TO_CREATE_ACCOUNTHOLDER	Account holder could not be created
UNABLE_TO_CREATE_INTEREST_TIER	Interest tier could not be created
UNABLE_TO_MODIFY_INTEREST_TIER	Interest tier could not be modified
UNABLE_TO_REMOVE_INTEREST_TIER	Interest tier could not be removed
UNBLOCK_ACCOUNT_FAILED	Account could not be un-blocked
UNKNOWN_ERROR	Unknown error.
UNKNOWN_URI	Provided URI is unknown to the server.
UNRESOLVED_REMITTANCE_ID_DOES_NOT_EXIST	Unresolved Remittance Id does not exist in dbase
UNSUPPORTED_FINANCIAL_TRANSACTION_ROLE	Unsupported financial transaction role
UNSUSPEND_CREDENTIAL_COUNTER_THRESHOLD_REACHED	The maximum attempts of credential unsuspend for the given account holder has been reached
USER_ID_NOT_ALLOWED_IN_PASSWORD	Using user id in password is not allowed
VALIDATION_ERROR	The data provided in the request failed the validation check
VOUCHER_DATE_OF_BIRTH_SET	Voucher date of birth has been set
VOUCHER_EXPIRED	Voucher has expired
VOUCHER_EXTERNALID_SET	Voucher external id has been set
VOUCHER_FIRSTNAME_SET	Voucher first name has been set
VOUCHER_NOT_ACTIVE	Voucher is no longer active
VOUCHER_NOT_FOUND	Voucher with given identity could not be found in the system
VOUCHER_SECRET_CONTAINS_EXCLUDED_LETTERS	Voucher secret contains letters that has been excluded through system configuration
VOUCHER_SECRET_CONTAINS_ILLEGAL_CHARACTERS	Voucher secret contains characters that is not legal according to system configuration
VOUCHER_SECRET_CONTAINS_NONASCII_CHARACTERS	Voucher secret contains non-ASCII characters
VOUCHER_SECRET_DIGITS_NOT_ALLOWED	Voucher secret can not contain digits
VOUCHER_SECRET_INVALID	Voucher secret is not correct
VOUCHER_SECRET_IS_EMPTY	Voucher secret element is empty
VOUCHER_SECRET_LETTERS_NOT_ALLOWED	Voucher secret can not contain letters
VOUCHER_SECRET_SYSTEM_CONFIGURATION_ERROR	The voucher secret rules has been incorrectly configured in the system
VOUCHER_SECRET_TOO_LONG	Voucher secret is longer than system configured maximum length
VOUCHER_SECRET_TOO_MANY_FAILED_REDEEM_ATTEMPTS	Too many redeem voucher attempts with incorrect secret has been performed
VOUCHER_SECRET_TOO_SHORT	Voucher secret is shorter than system configured minimum length
VOUCHER_SURNAME_SET	Voucher surname has been set
VOUCHER_TYPE_NOT_SUPPORTED	Voucher type not supported
WRONG	

